Light

GCSE Computer Science — Data Representation: Images and Sound (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a pixel?

- **A.** The smallest addressable element of a bitmap image
- **B.** A type of image file format
- **C.** The amount of memory used to store an image
- **D.** A measurement of screen brightness

Q2 (1 mark)

What does the term 'resolution' mean when describing a digital image?

- **A.** The file size of the image in bytes

- ○ **B.** The number of colours used in the image
- ○ **C.** The dimensions of the image expressed as width × height in pixels
- ○ **D.** The brightness level of the image

Q3 (1 mark)

What is colour depth?

- ○ **A.** The physical size of an image when printed
- ○ **B.** The number of bits used to represent the colour of each pixel
- ○ **C.** The total number of pixels in an image
- ○ **D.** The contrast between light and dark areas

Q4 (1 mark)

What is sampling in the context of digital sound?

- ○ **A.** Measuring the amplitude of a sound wave at regular intervals
- ○ **B.** Recording a short preview of a song
- ○ **C.** Converting digital audio back to analogue
- ○ **D.** Compressing audio files to reduce size

Q5 (1 mark)

What is sample rate?

- ○ **A.** The number of samples taken per second, measured in Hertz (Hz)
- ○ **B.** The volume level of a recording
- ○ **C.** The number of bits used per sample
- ○ **D.** The total length of an audio file

Q6 (1 mark)

In digital sound, what does bit depth determine?

- ○ **A.** How many different amplitude levels can be recorded for each sample
- ○ **B.** How long the audio recording can be
- ○ **C.** How many samples are taken per second
- ○ **D.** The frequency range of the audio

Q7 (1 mark)

An image uses a colour depth of 8 bits per pixel. How many different colours can be represented?

- ○ **A.** 8
- ○ **B.** 16
- ○ **C.** 64

- ○ **D.** 256

Q8 (1 mark)

Which two factors directly affect the file size of a bitmap image?

- ○ **A.** Brightness and contrast
- ○ **B.** Resolution and colour depth
- ○ **C.** File format and compression level
- ○ **D.** Screen size and refresh rate

Q9 (1 mark)

Which three factors affect the file size of a digital sound recording?

- ○ **A.** Bass, treble, and balance
- ○ **B.** Volume, pitch, and tempo
- ○ **C.** Frequency, wavelength, and speed
- ○ **D.** Sample rate, bit depth, and duration

Q10 (1 mark)

What is metadata in the context of a digital image?

- ○ **A.** The pixel colours stored in the image

- ○ **B.** Additional data stored with the image such as dimensions, date taken, and camera settings
- ○ **C.** A compressed version of the image
- ○ **D.** The file extension of the image

Q11 (1 mark)

What happens to an image when its resolution is increased?

- ○ **A.** The image contains more pixels and can show finer detail
- ○ **B.** The image uses fewer colours
- ○ **C.** The file size decreases
- ○ **D.** The image becomes darker

Q12 (1 mark)

Why does a higher sample rate generally produce better quality audio?

- ○ **A.** It captures more measurements of the sound wave per second, creating a more accurate representation
- ○ **B.** It makes the audio louder
- ○ **C.** It uses fewer bits per sample
- ○ **D.** It compresses the audio more efficiently

Q13 (2 marks)

An image has a resolution of 800 × 600 pixels and uses 24-bit colour depth.
Calculate the file size of this image in bytes.

- ○ **A.** 480,000 bytes
- ○ **B.** 1,440,000 bytes
- ○ **C.** 11,520,000 bytes
- ○ **D.** 14,400,000 bytes

Q14 (2 marks)

A sound file is recorded with a sample rate of 44,100 Hz, a bit depth of 16 bits, and lasts for 3 seconds.
Calculate the file size in bytes (assume mono audio).

- ○ **A.** 132,300 bytes
- ○ **B.** 16,934,400 bytes
- ○ **C.** 2,116,800 bytes
- ○ **D.** 264,600 bytes

Q15 (2 marks)

A graphic designer increases an image's colour depth from 8 bits to 24 bits per pixel.
Describe the effect on the image quality and file size.

- ○ **A.** Quality decreases and file size decreases

- ○ **B.** Quality increases and file size stays the same
- ○ **C.** Quality increases (from 256 to over 16 million colours) and file size triples
- ○ **D.** Quality stays the same and file size doubles

Q16 (2 marks)

Increasing the bit depth of a sound recording from 8-bit to 16-bit has which effects?

- ○ **A.** Doubles the sample rate and halves the file size
- ○ **B.** Increases the number of amplitude levels from 256 to 65,536 and doubles the file size
- ○ **C.** Makes the recording shorter and reduces quality
- ○ **D.** Increases the frequency range that can be captured

Q17 (2 marks)

A website designer needs to reduce the file size of an image. The current image is 2000 × 1500 pixels with 24-bit colour.
Which change would most significantly reduce file size while maintaining colour accuracy?

- ○ **A.** Increase the resolution to 4000 × 3000 pixels
- ○ **B.** Add more metadata to the image
- ○ **C.** Increase the colour depth to 32 bits
- ○ **D.** Reduce the resolution to 1000 × 750 pixels

Q18 (2 marks)

The human ear can typically hear sounds up to approximately 20,000 Hz. CD audio uses a sample rate of 44,100 Hz.
Explain why this sample rate is sufficient.

- ○ **A.** 44,100 is simply a convenient number chosen by manufacturers
- ○ **B.** CD audio doesn't actually capture the full range of human hearing
- ○ **C.** Higher sample rates would make the audio too loud for human ears
- ○ **D.** The sample rate must be at least twice the highest frequency to be captured, so 44,100 Hz can accurately capture frequencies up to about 22,050 Hz

Q19 (2 marks)

A small black and white image uses 1-bit colour depth and has a resolution of 8 × 8 pixels.
How much storage does the image data require?

- ○ **A.** 8 bits (1 byte)
- ○ **B.** 16 bits (2 bytes)
- ○ **C.** 64 bits (8 bytes)
- ○ **D.** 128 bits (16 bytes)

Q20 (2 marks)

Explain why digital audio is an approximation of the original analogue sound wave.

- ○ **A.** Digital audio is actually more accurate than analogue sound
- ○ **B.** Digital audio adds extra frequencies not present in the original
- ○ **C.** Analogue sound waves cannot be converted to digital format
- ○ **D.** Sampling only captures amplitude at discrete intervals, missing information between samples, and bit depth limits amplitude precision

Q21 (2 marks)

A photographer saves an image in JPEG format. The file size is much smaller than the original bitmap.
What type of compression has been applied and what is a potential drawback?

- ○ **A.** Lossless compression; no drawbacks as all data is preserved
- ○ **B.** Lossless compression; the image can only be viewed on certain devices
- ○ **C.** No compression; JPEG files are naturally smaller
- ○ **D.** Lossy compression; some image data is permanently discarded, which may reduce quality

Q22 (2 marks)

A stereo audio file (2 channels) is recorded at 48,000 Hz sample rate, 24-bit depth, for 10 seconds.
Calculate the file size in megabytes (MB). Use 1 MB = 1,000,000 bytes.

- ○ **A.** 1.44 MB
- ○ **B.** 2.88 MB
- ○ **C.** 11.52 MB

- ○ **D.** 28.8 MB

Q23 (2 marks)

A game developer needs images that support transparency. They choose 32-bit colour depth (24 bits for colour + 8 bits for alpha/transparency).
An image is 400 × 300 pixels. What is the file size in kilobytes (KB)? Use 1 KB = 1,000 bytes.

- ○ **A.** 120 KB
- ○ **B.** 360 KB
- ○ **C.** 480 KB
- ○ **D.** 960 KB

Q24 (3 marks)

A photographer has a 12-megapixel camera (4000 × 3000 pixels) and needs to print a high-quality A4 poster. The printer requires at least 300 pixels per inch (PPI) for professional quality. An A4 page is approximately 11.7 × 8.3 inches.
Analyse whether the camera resolution is sufficient.

- ○ **A.** The resolution is insufficient because 12 megapixels is too low for any print
- ○ **B.** The resolution is exactly right with no pixels to spare
- ○ **C.** The resolution is sufficient because 4000 × 3000 pixels provides approximately 342 × 361 PPI when printed at A4, exceeding the 300 PPI requirement
- ○ **D.** The resolution is too high and will cause printing problems

Q25 (3 marks)

A music streaming service offers three quality levels:

- Low: 22,050 Hz, 8-bit
- Medium: 44,100 Hz, 16-bit
- High: 96,000 Hz, 24-bit

Analyse how much larger the high quality file would be compared to the low quality file for the same song duration.

- ○ **A.** 2 times larger
- ○ **B.** 6 times larger
- ○ **C.** 13 times larger (approximately)
- ○ **D.** 48 times larger

Q26 (3 marks)

A company logo needs to be used on business cards (small), letterheads (medium), and billboards (very large).
Analyse whether a bitmap or vector format would be more suitable and explain why.

- ○ **A.** Vector format is better because it uses mathematical descriptions of shapes that can be scaled to any size without losing quality, while bitmaps would become pixelated when enlarged
- ○ **B.** Bitmap format is better because it can store more colours
- ○ **C.** Neither format is suitable for logos
- ○ **D.** Bitmap format is better because vector images cannot be printed

Q27 (3 marks)

An uncompressed CD-quality audio file (44,100 Hz, 16-bit, stereo) for a 4-minute song is approximately 40 MB. The same song as a compressed MP3 file is 4 MB.
Analyse the compression ratio and discuss the trade-offs involved.

- ○ **A.** The compression ratio is 10:1; MP3 achieves this by removing all data above 15,000 Hz
- ○ **B.** The compression ratio is 40:1; this is achieved without any quality loss
- ○ **C.** The compression ratio is 4:1; MP3 uses lossless compression that preserves all original data
- ○ **D.** The compression ratio is 10:1; MP3 uses lossy compression that removes audio data humans are less likely to hear (psychoacoustic model), reducing file size but permanently discarding some audio information

Q28 (3 marks)

A medical imaging system displays X-ray images using 16-bit greyscale (65,536 shades of grey) rather than standard 8-bit greyscale (256 shades).
Analyse why this higher bit depth is important for medical applications.

- ○ **A.** 16-bit images load faster on medical displays
- ○ **B.** 16-bit greyscale allows doctors to distinguish 256× more subtle variations in tissue density, enabling detection of small abnormalities that might be invisible in 8-bit; the extra precision can be critical for accurate diagnosis
- ○ **C.** 8-bit greyscale would be sufficient but hospitals use 16-bit for marketing purposes
- ○ **D.** 16-bit images use less storage space than 8-bit images

Q29 (3 marks)

An audio engineer notices 'quantisation noise' — a faint hissing sound — in an 8-bit recording that isn't present in the original analogue source.
Analyse what causes this noise and how it can be reduced.

- ○ **A.** Quantisation noise is caused by electrical interference and can be reduced by using shielded cables
- ○ **B.** Quantisation noise cannot be reduced — it is inherent to all digital audio regardless of settings
- ○ **C.** Quantisation noise is caused by sampling too frequently and can be reduced by lowering the sample rate
- ○ **D.** Quantisation noise occurs because 8-bit only allows 256 amplitude levels, so actual values must be rounded to the nearest available level; the small rounding errors create audible noise, which can be reduced by using higher bit depth (e.g., 16-bit or 24-bit)

Q30 (3 marks)

A smartphone has 64 GB of storage. A user wants to know whether they can store more high-resolution photos (5000 × 4000 pixels, 24-bit colour, approximately 60 MB each) or CD-quality songs (44,100 Hz, 16-bit, stereo, 4 minutes each, approximately 40 MB each).
Analyse and compare the storage options.

- ○ **A.** They can store approximately the same number of photos and songs because the file sizes are similar
- ○ **B.** They can store approximately 1,067 photos or 1,600 songs; despite photos being larger individually, the user can store more songs because audio compression makes most music files much smaller than 40 MB in practice
- ○ **C.** They can store more photos because images compress better than audio

○ **D.** They cannot store either format because 64 GB is insufficient for media files

Total: **55** marks • Time: **30 min**

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